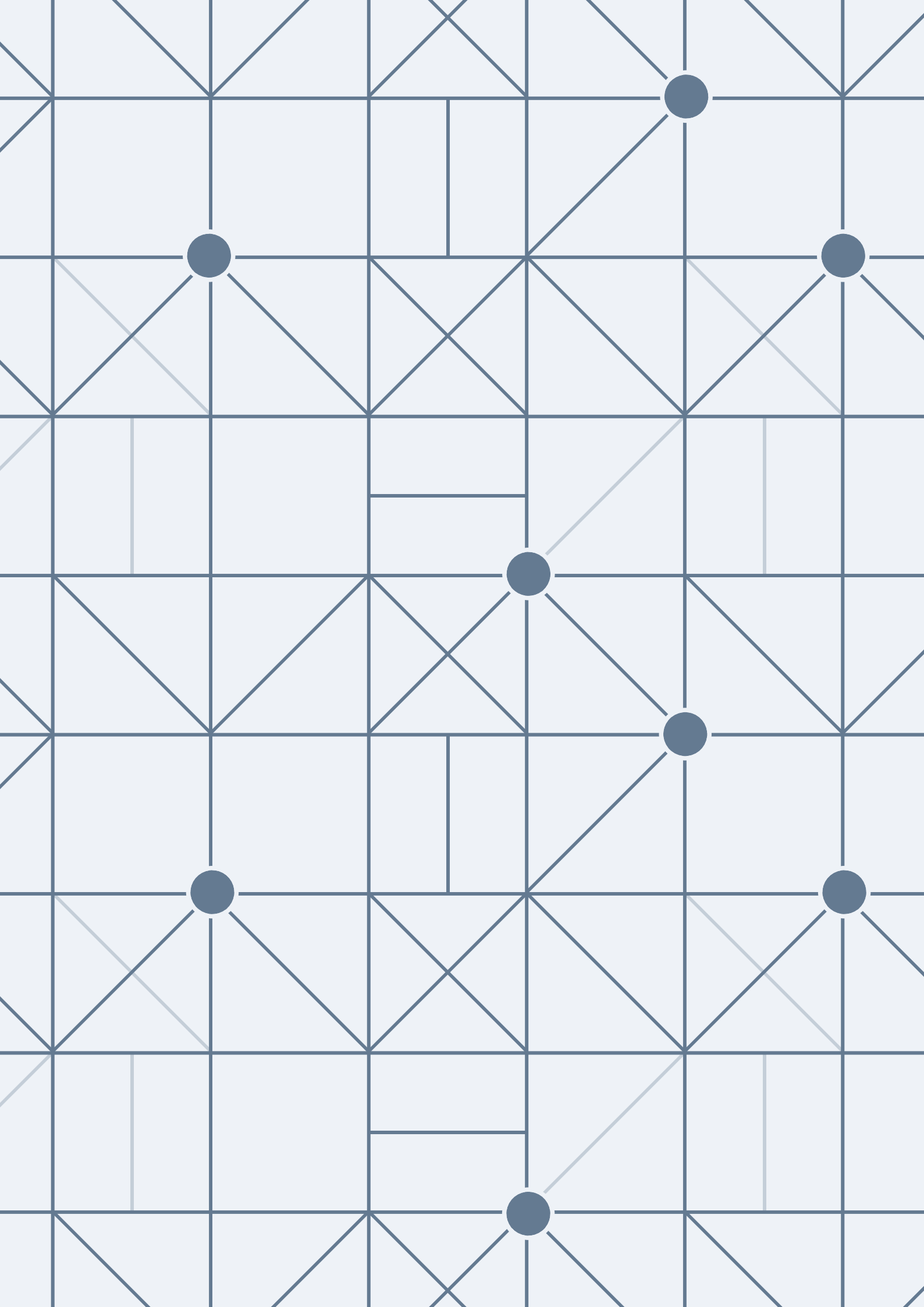




$$H = -\frac{1}{2} \sum_{i,j} \frac{q_i q_j}{r_{ij}} \Rightarrow \text{Act interacting many body}$$

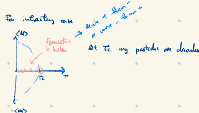
classical system

or

or

or

or



$$Z = \int e^{-\beta H} d\mathbf{r} d\mathbf{p} = \int e^{-\beta H} d\mathbf{r} d\mathbf{p}$$

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$$U = \left\langle \frac{\partial \ln Z}{\partial \beta} \right\rangle = \left\langle \frac{\partial}{\partial \beta} \left( \ln \int e^{-\beta H} d\mathbf{r} d\mathbf{p} \right) \right\rangle = \left\langle \frac{\partial}{\partial \beta} \left( \ln \int e^{-\beta H} d\mathbf{r} d\mathbf{p} \right) \right\rangle$$

$$U = \langle H \rangle = \frac{1}{Z} \int H e^{-\beta H} d\mathbf{r} d\mathbf{p} = \frac{1}{Z} \int H e^{-\beta H} d\mathbf{r} d\mathbf{p}$$

$$U = \langle H \rangle = \frac{1}{Z} \int H e^{-\beta H} d\mathbf{r} d\mathbf{p} = \frac{1}{Z} \int H e^{-\beta H} d\mathbf{r} d\mathbf{p}$$

1d system

System with  $N$  identical particles

Hamiltonian

Hamiltonian is given by

$$H = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$$

$H$  is a function of  $N$  coordinates and  $N$  momenta

$$H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$$

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Canonical ensemble:  $N$  is large,  $N \rightarrow \infty$

$$H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$$

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Hamiltonian

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What happens if the particles are in the same position, energy and momentum?  $H \rightarrow \infty$

First step:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$   $\Rightarrow$  The particles are in the same position, energy and momentum.  $\Rightarrow$  The particles are in the same position, energy and momentum.  $\Rightarrow$  The particles are in the same position, energy and momentum.

Second step:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$   $\Rightarrow$  The particles are in the same position, energy and momentum.  $\Rightarrow$  The particles are in the same position, energy and momentum.  $\Rightarrow$  The particles are in the same position, energy and momentum.

Examples:

Example 1:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$

Example 2:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$

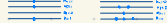
Example 3:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$

Example 4:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$

Example 5:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$

Example 6:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$

Example 7:  $H(x_1, \dots, x_N, p_1, \dots, p_N) = \sum_{i=1}^N \left( \frac{p_i^2}{2m} + V(x_i) \right)$



$$Z = \int e^{-\beta H} d\mathbf{r} d\mathbf{p} = \int e^{-\beta H} d\mathbf{r} d\mathbf{p}$$

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